Alyssa Dang

adang2@andrew.cmu.edu | linkedin.com/in/alyssakdang | github.com/alyssahui | alyssahui.github.io

EDUCATION

Carnegie Mellon University, Pittsburgh, PA *Expected May 2028*

B.S. in Human-Computer Interaction + Cognitive Science (Machine Learning) | GPA: 4.0/4.0

- Relevant Coursework: Designing Human-Centered Software, Computational Neuroscience, Imperative Computation, Cognitive Psychology
- Activities: Scottylabs Design, Game Creation Society, Neuroscience Student Advisory Council, CMU Mock Trial

RELEVANT EXPERIENCE

Product & UI/UX Designer, CravingChef (iOS App Store, Food AI Agent) *Jul 2026 – Aug 2026*

- Synthesized user feedback and behavioral analytics into a Figma overhaul of the main feed, launching custom in-feed recipe "interruption cards" and boosting engagement with user-uploaded content
- Owned low-to-high fidelity UI flows and prototypes for a "Find Places" restaurant locator, public recipe upload hub, user reviews, and recipe iteration toggles; drove engineering handoff for live App Store releases

Research Assistant, Laboratory of Computational Perception *Aug 2026 – Present*

- Refining pipeline to reliably simulate movement and muscle kinematics (MuJoCo, MyoSuite), derive muscle spindle firing, and train CNNs to predict proprioceptive illusions and biases, verified against human data

Project Lead, Honey Haven, CMU Game Creation Society *Jan 2026 – Apr 2026*

- Won Game Creation Society Gold Award (2026) leading a cross-functional team of designers, programmers, and artists to pitch, direct, and ship a psychological horror visual novel in one semester

RESEARCH & TECHNICAL PROJECTS

V1 Neural Decoding & Systems Identification, Computational Neuroscience Lab *Apr 2026*

- Analyzed mouse V1 Neuropixel recordings (Allen Institute/DANDI) using tuning curves, receptive field mapping, and Linear-Nonlinear-Poisson (LNP) model validation; applied PCA via SVD, with 6 components capturing 95% of population variance
- Built OLS linear regression decoders with circular target encodings (80/20 train/test split), decoding orientation more accurately than direction (31° vs. 67° mean test error)

LTP/LTD Circuit Simulation, Computational Neuroscience Lab *Feb 2026*

- Built a Leaky Integrate-and-Fire circuit model in Python (NumPy, Matplotlib) simulating AMPA/NMDA kinetics, implementing calcium-threshold STDP learning rules and validating against published literature

NBA Outcome Prediction & Optimization Model, ML & Probabilistic Modeling *Apr 2026*

- Modeled NBA home win probabilities in R with Random Forest classifiers; reached a 0.102 Brier score via 5-fold cross-validation and quantified prediction uncertainty to guide expected-value decisions

DESIGN PROJECTS

SideQuest (Mobile App) — Product & UI/UX Design *Sep 2026*

- Designed features and flow for a HackCMU app turning daily life into an open-world RPG with self-created "quests" and Kalshi-style "challenges"; designed quest and bet screens and produced the demo video

2529 Films Studio Website — UI/UX Design & Development *Jul 2026*

- Increased studio engagement by 50% by designing and deploying a responsive website; built branding, layout grids, typography systems, and booking flows in Figma before front-end execution

Sir Jester (Retro Mobile Game) — UI/UX Design *Feb 2026*

- Designed minimalist, offline-playable mobile UI/UX within a strict 15KB memory constraint during a 24-hour hackathon

SKILLS

Design Methods: User Research, Usability Testing, Information Architecture, Wireframing, Prototyping, Design Systems

Technical & Research: Python, PyTorch, NumPy, Matplotlib, R, C, HTML/CSS, JavaScript, Git, Jupyter, PCA/SVD, Regression Decoders, Computational Modeling, Cross-Validation

Tools: Figma, Adobe Creative Suite, Final Cut Pro, CapCut, Google Workspace